



Measurement ...: Si wafer testing

Measurement Exercise for Physicist-Engineer Students

Abstract. Charge carrier recombination lifetime is one of the most important electrical parameters for semiconductor materials, especially in photovoltaic applications. The bulk recombination rate within a silicon wafer is determined by the initial crystal quality and the density of defects and impurities. Therefore, accurate measurement of the recombination lifetime provides valuable information about the crystal's purity. In industrial production, recombination lifetime measurement is a standard method for quality control. This measurement practice consists of two parts. First, the iron concentration in a silicon wafer (suitable for semiconductor industry use) is determined by performing recombination lifetime measurements in different sample states. Second, the effect of charge carrier diffusion processes on recombination lifetime is investigated in silicon wafers used for solar cells.

Warning. The measurement systems used during the practice contain high intensity light sources, flash lamp and lasers. These sources can emit visible and invisible light that may be harmful to human eyes. During normal operation there is no risk of exposure to laser radiation.

The WT-2000 and WT-1200IL systems are designed to be completely safe when operated and maintained by properly trained personnel using procedures described by Semilab. For a more detailed overview of safety precautions, please refer to the attached device manuals and their Safety Precautions section.

During the measurements, always look out for your own and other people's safety. If you are not sure about something, feel free to ask for help from the responsible professors and assistants present the laboratory and always respect their warnings.

Theoretical summary

I

1 Recombination in silicon

The recombination lifetime is one of the most important physical parameters of a semiconductor material, as it may provide information about the presence of harmful defects and contaminants. Even in dark thermal equilibrium, charge carriers are continuously generated by thermal excitation and recombine with each other through various processes. If excess electron-hole pairs are generated in the semiconductor instantaneously, the recombination rate increases, and the decay of the charge carrier density can be characterized by the recombination lifetime. In more general terms, the recombination lifetime is defined as:

$$\tau = \frac{\Delta n}{R} \quad (\text{I.1.1})$$

where Δn is the excess minority carrier concentration or as it is called in solar cell terminology: injection level, and R is the recombination rate.

The charge carrier diffusion length refers to the average distance that a charge carrier can diffuse within a semiconductor material before recombination occurs. It is closely related to recombination lifetime:

$$L_D = \sqrt{D \cdot \tau}, \quad (\text{I.1.2})$$

where D is the diffusion coefficient of the minority charge carriers.

1.1 Recombination processes

The rate of most recombination processes depends on the charge carrier density. The exact dependence ($R(\Delta n)$) is determined by the number of charge carriers involved in a single recombination event. The three major recombination mechanisms are summarized and illustrated in Figure I.1.1. The type of a recombination event relates to the mechanism how the band gap equivalent energy is emitted during the recombination event.

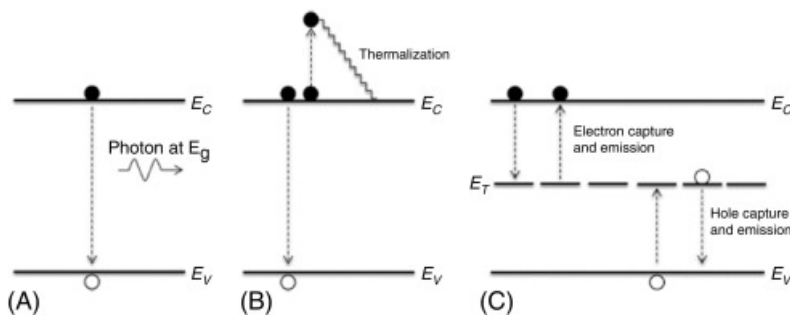


Figure I.1.1. The main charge carrier recombination processes in semiconductors. A) Radiative recombination, B) the Auger process, C) the Shockley-Read-Hall mechanism [1].

Typically, the different recombination processes occur simultaneously. In this case, the total recombination rate is the sum of individual contributions:

$$R_{total} = R_{rad} + R_{Auger} + R_{SRH} \quad (\text{I.1.3})$$

Then the effective carrier lifetime, τ_{eff} , can be expressed by

$$\frac{1}{\tau_{eff}} = \frac{1}{\tau_{rad}} + \frac{1}{\tau_{Auger}} + \frac{1}{\tau_{SRH}} \quad (\text{I.1.4})$$

In silicon samples for solar cells, the dominant recombination processes are typically defect related Shockley–Read–Hall (SRH) at low injection levels and Auger recombination at high injection levels as depicted in Figure I.1.2 with $\tau_{eff} = \tau_{bulk}$ as only bulk recombination processes were considered.

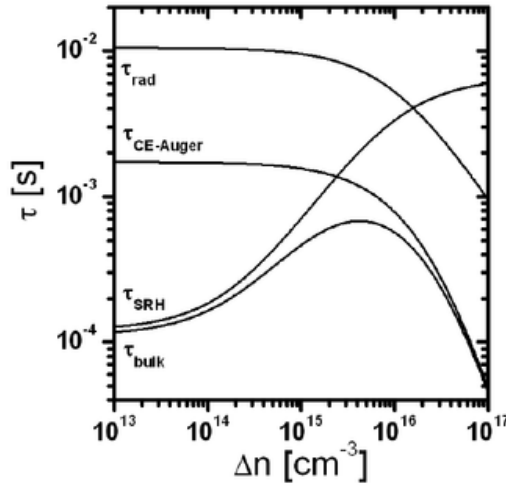


Figure I.1.2. The bulk recombination lifetime is dominated by different recombination mechanisms at low and high injection levels [2].

1.2 Surface recombination

Surface recombination refers to the process in which charge carriers (electrons and holes) recombine at the surface or interface of a semiconductor material, rather than in the bulk. This phenomenon is particularly important in semiconductor devices, such as solar cells, since their thickness is smaller than the diffusion length. Therefore, surfaces often significantly influence the overall device performance. The large variety of dangling bonds on the surface and other structural imperfections manifest in surface states, which have energy levels located within the band gap of the semiconductor. Therefore, their recombination properties can be described with the Shockley-Read-Hall model as well. However, instead of one discrete energy level, there is a spectrum of energy states at the surface. The practical approach to mathematically handle the surface recombination phenomenon in a universal way applies the parameter called surface recombination velocity S . Its definition is similar to the carrier lifetime integrating all recombination events corresponding to the surface:

$$S = \frac{R_{surf}}{\Delta n_{surf}} \quad (\text{I.1.5})$$

where Δn_{surf} is the surface excess carrier density and R_{surf} is the recombination rate on the surface. S has dimensions of m/s (the majority of the literature uses cm/s), which stems from the fact that R_{surf} is a surface quantity with dimensions of $\text{m}^{-2}\text{s}^{-1}$.

For the measurement of thin wafers with thickness $W < L_D$, a surface recombination lifetime, τ_{surf} can be defined:

$$\tau_{\text{surf}} \approx \frac{W}{2S} + \frac{W^2}{\pi^2 D}. \quad (\text{I.1.6})$$

where W is the thickness of the wafer and D is the diffusion coefficient.

In two limiting cases, the expression can be further simplified. First, in the case of passivated surface, which reduces the surface recombination velocity. Passivating the surface involves coating the sample with materials (such as silicon dioxide, silicon nitride, or organic layers) that reduce the density of surface states, thereby lowering the surface recombination velocity.

$$\tau_{\text{surf}}(S \rightarrow 0) = \frac{W}{2S}. \quad (\text{I.1.7})$$

While in the opposite case ($S \rightarrow \infty$):

$$\tau_{\text{surf}}(S \rightarrow \infty) = \frac{W^2}{\pi^2 D}. \quad (\text{I.1.8})$$

This is the so-called diffusion-limited case, typically observed in as-cut or polished surfaces, when the surface recombination is limited by the minority carrier diffusion process.

This expression enables us to categorize recombination processes according to the location where they occur:

$$\frac{1}{\tau_{\text{eff}}} = \frac{1}{\tau_{\text{bulk}}} + \frac{1}{\tau_{\text{surf}}}. \quad (\text{I.1.9})$$

2 Recombination lifetime measurement principles

Recombination lifetime is a local property of the silicon crystal. However, due to the movement of charge carriers, its effect on devices is not localized to where the recombination occurs. During carrier lifetime characterization, an apparent effective carrier lifetime (τ_{eff}) is measured, which in some cases provides direct information regarding the electrical performance of the final device (e.g. measurement on solar cell structures before metallization). In other cases, the measurement procedure must be adjusted or additional sample treatments have to be applied to obtain the information of interest. Such treatment is the surface passivation of the sample in order to obtain the carrier lifetime within the wafer.

To evaluate and understand the recombination dynamics in silicon materials and devices, several experimental techniques have been developed. They are essential for optimizing the material quality and the device performance, especially for photovoltaic applications. Technically, any carrier lifetime characterization technique is based on the same phenomenon: the injection of excess charge carriers, and the monitoring of their actual density Δn . To simplify the discussion, the average of the carrier density in the depth is taken, so $\Delta n_{\text{average}}(t) = \frac{\int_0^W \Delta n(x,t) dx}{W}$. Using the formulas for τ_{eff} in Equation I.1.9, the impact of surface recombination and diffusion process can be handled mathematically using a single τ_{eff} value. Therefore, we get a simple approximate formula for $\Delta n_{\text{av}}(t)$:

$$\frac{\partial \Delta n_{\text{av}}(t)}{\partial t} = G - R(\Delta n_{\text{av}}(t)) = G - \frac{\Delta n_{\text{av}}(t)}{\tau_{\text{eff}}(\Delta n)} \quad (\text{I.2.1})$$

where G is the generation rate of the optical injection and $R(\Delta n)$ is the total recombination rate of the carriers. Rearranging this equation, the general expression of the carrier lifetime

measurement can be determined:

$$\tau_{eff}(\Delta n_{av}) = \frac{\Delta n_{av}}{G - \frac{\partial \Delta n_{av}}{\partial t}}. \quad (\text{I.2.2})$$

From this point, to simplify the equations, $\Delta n := \Delta n_{av}$ always refers to the average injection level, while if the local carrier density needs to be applied, it will be indicated as $\Delta n(x)$.

There are two main methods to measure the charge carrier lifetime. The steady state (SS) and the transient method, which is realized using the photoconductance decay (PCD) technique in practice.

In the SS method, the charge carriers are excited optically – with laser in our case – until the injection level reaches the steady state. In the steady state case $\frac{\partial \Delta n}{\partial t} = 0$, therefore, equation I.2.2 simplifies as:

$$\tau(\Delta n) = \frac{\Delta n}{G} \quad (\text{I.2.3})$$

The PCD method starts with a laser – or other optical – excitation. After instantaneously switching off the excitation laser, injection level and the measured signal relaxes back to the equilibrium state. During the decay, the excitation is zero ($G = 0$) and the lifetime can be determined in the function of Δn :

$$\tau(\Delta n) = -\frac{\Delta n}{\frac{\partial \Delta n}{\partial t}} \quad (\text{I.2.4})$$

2.1 μ PCD measurement

The microwave induced photoconductance decay (μ PCD) method is the most common way of measuring minority carrier lifetime in semiconductors. This method excels due to its reliability, good reproducibility, and the short measurement time that permits recording lifetime maps with high resolution.

The pulse of an infrared semiconductor laser generates free electron-hole pairs under the illuminated area on the sample (Figure I.2.1). Since the penetration depth of the 904 nm wavelength light is about 30 μm in silicon, the free carriers are generated close to the front surface of the sample.

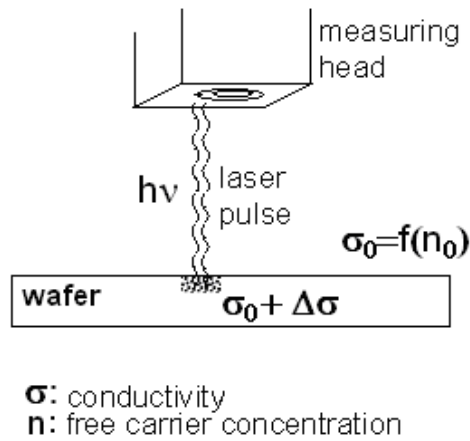


Figure I.2.1. Laser excitation

Since the free electrons and holes recombine, their concentration and so the conduc-

tivity of the sample decreases after the excitation. The decaying conductivity can be monitored by detecting the microwave reflectivity, because the reflected microwave power depends on the conductivity of the sample (Figure I.2.2). It is measured as a function of time.

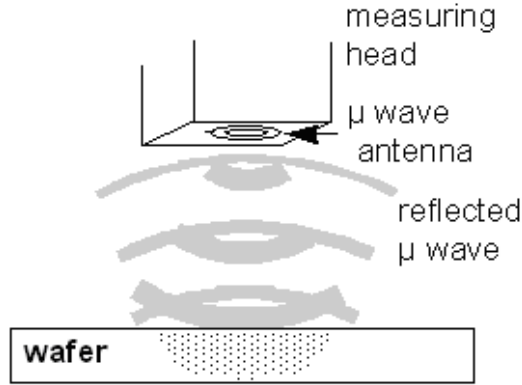


Figure I.2.2. Detection by microwave

The ring-shaped microwave antenna is operated near to its resonance frequency, in the range of 10.0 GHz – 10.5 GHz. The sample and the antenna above it, compose a resonator, that's resonance curve can be seen on Figure I.2.3, red line. As the conductivity of the sample increases due to the excess carriers generated by the illumination, the resonance curve shifts and gets deformed (Figure I.2.3, blue line). As the excess carriers recombine, the conductivity of the sample returns to equilibrium, and the resonance curve gets back to its original state. During measurement, the microwave frequency is set to a predefined frequency (vertical, green, broken line on Figure I.2.3). As the curve gets back to the original state, the measured signal of the antenna changes.

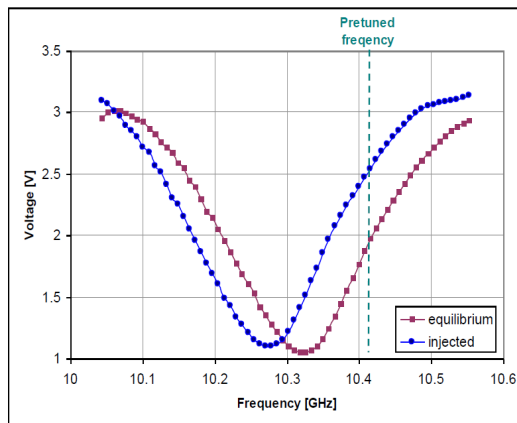


Figure I.2.3. Resonance curves of the microwave probe with a sample underneath. When the sample is in equilibrium, the curve is the red line. When the sample is injected, excess carriers are generated by the illumination, the conductivity and reflectance is changed, the curve is the blue line. The vertical green line shows the frequency used during the measurement, which is tuned so that the change of the signal is maximal.

For this technique, the measured signal is assumed to be proportional to the number of excess carriers, and the recombination rate is assumed to be constant. Therefore, the recombination time is determined by fitting an exponential decay function to the recorded

decaying signal. (Figure I.2.4):

$$\Delta n = \text{const} \cdot \exp(-t/\tau_{eff}) \quad (\text{I.2.5})$$

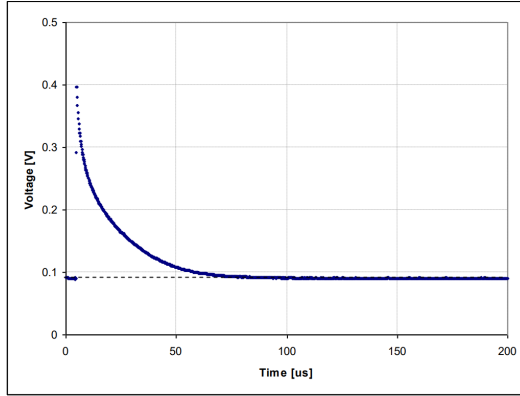


Figure I.2.4. Transient, measured on a sample. As the carriers return to equilibrium exponentially, the signal changes accordingly.

The μ PCD measurement technique will be used in the WT-2000 system, which is capable to perform high resolution carrier lifetime maps.

2.2 Iron contamination measurement

The most often detected contaminant in silicon is iron. In p-type silicon iron is present as an iron-boron (Fe-B) pair. This pair is a very effective recombination center. The Fe-B pair dissociates to interstitial iron (Fe_i) and boron after heat treatment (200°C for 10 minutes with fast quenching after the treatment) or upon high-intensity illumination. This unique behavior enables its identification because the recombination efficiency of the Fe-B pair and that of the Fe_i is different. At high injection level, which is used in the μ PCD technique, the Fe-B pair causes lower lifetime than the Fe_i , so during the dissociation process, the lifetime increases. This can be recorded by in situ lifetime measurement (see Figure I.2.5). Once iron is separated from boron an association process takes place even at room temperature which results in decreasing lifetime. Its time constant can be some hours or days depending on the boron concentration (i.e. the resistivity) of the wafer under test.

Figure I.2.5 shows a typical sequence. The sample was illuminated with fast repeated laser pulses to break the Fe-B pairs. After switching the laser on, the lifetime increases quickly at the beginning then more slowly to a saturation value. After switching the illumination off, the interstitial iron again forms a pair with boron, and the lifetime decreases with a given time constant to the original value. The Fe concentration can be calculated from the lifetime measured before and after the dissociation with the following formula:

$$N_{Fe} = C_{\mu PCD} \left(\frac{1}{\tau_{before}} - \frac{1}{\tau_{after}} \right) \quad (\text{I.2.6})$$

where $C_{\mu PCD} [\mu\text{s}/\text{cm}^3]$ is the iron constant. The association kinetics of Fe-B pairs is given by [7]

$$R = \frac{e^2}{\varepsilon \varepsilon_0 k_B T} N_B D_0 \exp \left(-\frac{E_{mig}}{k_B T} \right), \quad (\text{I.2.7})$$

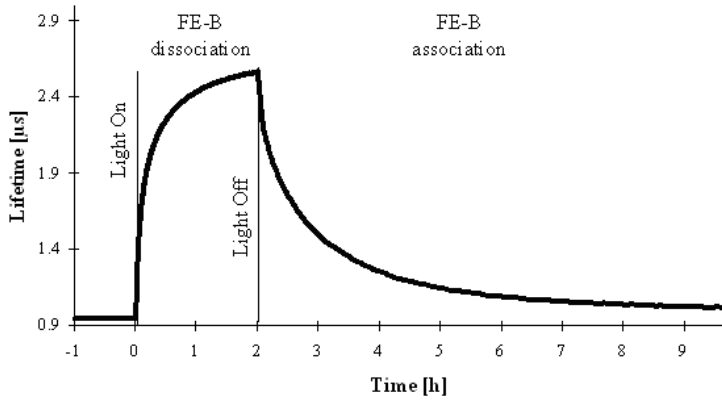


Figure I.2.5. Fe-B dissociation and association over time

where ϵ is the dielectric constant of silicon, e is the unit charge, k_B is the Boltzmann's constant, T is the absolute temperature, N_B is the concentration of boron, D_0 is the iron diffusion constant and $E_{mig} = 0.64 \pm 0.01$ eV [8] is the activation energy of Fe_i^+ migration.

2.3 ePCD measurement

The eddy-current based Photoconductance Decay (ePCD) measurement is developed for investigating thick silicon samples. For this purpose, a longer wavelength infrared laser is used with deeper photogeneration. It also applies light pulses to generate free electron-hole pairs deep in the bulk under the illuminated area on the sample (Figure I.2.6/a).

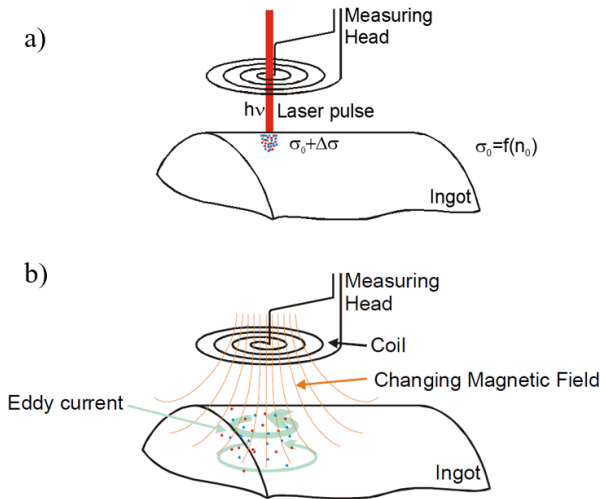


Figure I.2.6. The operating principle of eddy current measurements.

For the accurate detection of the deeply generated charge carriers, the eddy-current technique is used (Figure I.2.6/b). This is based on the interaction between an alternating magnetic field and free charges (electrons or holes) in a conductive material. An oscillating magnetic field near the material surface induces circulating currents (eddy currents) within the material. These currents generate their own magnetic fields, which oppose the original magnetic field according to Lenz's law. The magnitude of eddy currents and their interac-

tion with the magnetic field is influenced by the electrical conductivity (or resistivity) of the material as well as its thickness and other electromagnetic properties.

In Semilab systems, an alternating current (AC) is passed through a coil placed near the sample, and the response of the eddy currents is measured through changes in the impedance of the coil. The depth at which the electromagnetic field penetrates and induces eddy currents depends on the skin depth, determined by the electrical resistivity of the material and the frequency of the applied magnetic field:

$$\delta = \sqrt{\frac{2\rho}{\mu\omega}}, \quad (\text{I.2.8})$$

where ρ is the resistivity of the material, μ is the permeability (for silicon, it is almost equal to vacuum permeability), and ω is the angular frequency of the applied field. The resistivity of silicon typically used in solar cell applications ($\approx 1 \Omega\text{cm}$) results in a skin depth of around 1 cm, which implies that the entire depth of thin wafers ($W < 200 \mu\text{cm}$) can be properly sensed, while it is questionable for thick samples, ingots.

2.4 Evaluation of ePCD curves

While certain applications, such as rapid silicon wafer mapping, may utilize a simplified approach of extracting a single lifetime value through exponential function fitting, more sophisticated analyses of recombination properties necessitate the determination of the excess-carrier-concentration-dependent recombination lifetime function, $\tau(\Delta n)$.

The importance of obtaining the $\tau(\Delta n)$ function is particularly pronounced when examining exotic sample types, where recombination lifetime can exhibit order-of-magnitude variations during decay, primarily due to charge carrier trapping phenomena. This function provides crucial insights into the complex recombination dynamics occurring within the material.

The objective of this evaluation procedure is to derive the $\tau(\Delta n)$ function from the observed decay curve. In the standard PCD measurement protocol, multiple sequential excitation pulses are typically employed to enhance measurement accuracy. The subsequent signal decay curves are averaged, resulting in a significant improvement in the signal-to-noise ratio. The number of averages is typically optimized based on the measured carrier lifetime and the signal magnitude to balance the measurement time and the accuracy. The equilibrium state signal is recorded prior to excitation as it provides the conductivity and doping level.

The determination of carrier lifetime necessitates the numerical differentiation of the decay curve. With regard of the sensitivity of numerical differentiation to noise, the elimination of external disturbances is crucial for an accurate evaluation (e.g. blue triangles in Figure I.2.7a.).

The subsequent phase in the evaluation process involves the computation of excess charge carrier concentration (Δn) decay from the averaged signals as it is presented in Figure I.2.7b. For eddy current measurement techniques, the measured signal can be easily calibrated to the sheet conductance (σ_s) of silicon wafers. To avoid the inaccuracies, calibration wafers of similar thickness to the sample under investigation need to be employed.

The standard formula to calculate the density of excess carriers from the sheet con-

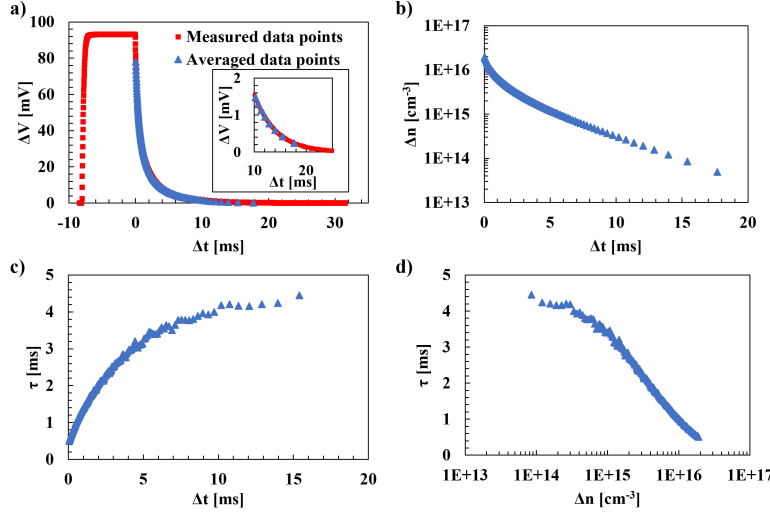


Figure I.2.7. a) measured signal during an ePCD measurement, b) $\Delta n(t)$ curve of the decay, c) $\tau(t)$ curve of the decay, d) $\tau(\Delta n)$ lifetime curve of the ePCD measurement.

ductance is:

$$\Delta n = \frac{\Delta \sigma_s}{W \cdot e \cdot \mu_{\text{sum}}(\Delta n, N_{\text{dop}})}, \quad (\text{I.2.9})$$

where e is the electron charge, $\Delta \sigma_s$ is the excess sheet conductance, and $\mu_{\text{sum}} = \mu_e + \mu_h$ is the sum of electron and hole mobilities.

The charge carrier mobility is not a constant parameter but rather a function of the dopant atom concentration and the excess charge carrier density. Consequently, an iterative calculation is used to accurately determine Δn . The mobility shift of dopant-related free carriers also contributes to the observed conductance in the injected case. Considering this effect, the expression for Δn is modified as follows:

$$\Delta n = \frac{\sigma_s/W - N_{\text{dop}} \cdot e \cdot \mu_{\text{maj}}(N_{\text{dop}}, \Delta n)}{e \cdot \mu_{\text{sum}}(N_{\text{dop}}, \Delta n)}, \quad (\text{I.2.10})$$

where N_{dop} is the doping concentration and μ_{maj} is the mobility of majority charge carriers.

Numerous theoretical and empirical models are available to determine the mobility of electrons and holes in silicon. Among these, Klaassen's semi-empirical model is used, which not only accounts for carrier concentration but also incorporates temperature dependence, providing a more accurate representation of charge carrier mobility across a wide range of operating conditions. For the sake of ease, a data sheet is provided with the corresponding mobility.

The concluding phase of the evaluation process involves the numerical differentiation of the $\Delta n(t)$ curve, followed by the computation of the recombination lifetime corresponding to each data point based on Equation I.2.4 (Figure I.2.7c and d).

Measurement tools

II

WT-2000

The WT-2000 system is a platform that can be equipped with a variety of measuring options. This allows characterization of silicon wafers in diverse ways. It is a useful tool for incoming wafer inspection, quality control and process monitoring in the wafer and IC manufacturing processes. In this measurement exercise the prepared measurement option is the μ PCD. The system contains the below mentioned parts:

1. WT-2000 main unit with scanning capability
2. Sample stage
3. Indexer for automatic wafer handling from a wafer cassette
4. Two computers, including a Measuring computer (System 1, DOS) and a User computer (System2, Windows)
5. Peripherals including keyboard, mouse, LCD monitor
6. μ PCD head for lifetime and Fe concentration measurements
7. Flash unit for whole wafer, providing high intensity illumination (for Fe measurement)

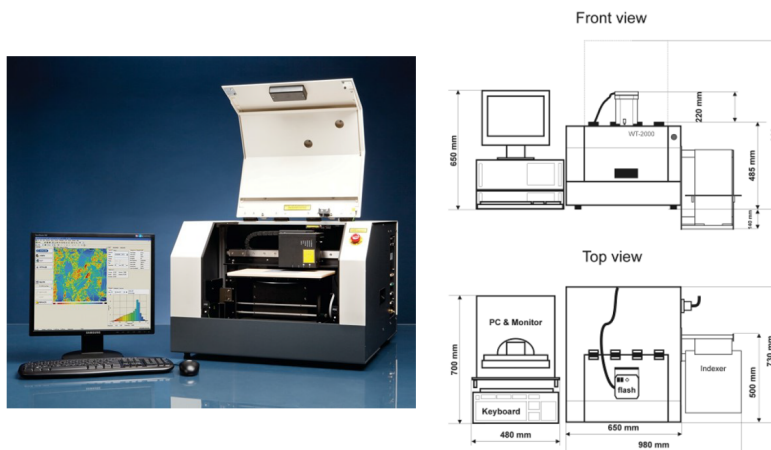


Figure II.2.8. Semilab WT-2000 charge carrier lifetime mapping tool.

WT-1200IL

The WT-1200IL is a single-point non-contact carrier lifetime measuring device capable of monitoring defects and contamination both in the bulk and in the surface region of silicon ingots. It is primarily optimized for the investigation and quality control of ingots, but can also be used with wafers and solar cells.

1. Laser unit box with measuring head
2. Industrial PC (Windows operation system)
3. Peripherals (keyboard, mouse, LCD monitor)
4. μ PCD cable
5. Serial cable
6. Power cable
7. Main cord for industrial PC
8. Main cord for laser control box
9. Main cord for monitor



Figure II.2.9. WT-1200IL hand-held recombination lifetime measurement tool

This device is capable to perform both ePCD and SS measurements on wafers and ingots.

3 Measurement tasks

3.1 Measurement of the dynamics of iron contamination in silicon

3.1.1 Investigation of the association of Fe-B pairs

The iron concentration is proportional to the difference of reciprocal recombination lifetime between the FeB and Fe_i states (eq. 1.2.6). In this first task, determine the proportionality factor (iron constant) on a sample with known [Fe] (B2).

Task 1. Load the measurable wafer (B2) into the WT-2000 system and run a mapping measurement. Use the "Autosetting" function to set suitable measuring parameters and save the results. Use 2mm measurement raster. Use these settings when further examining this sample.

Task 2. Perform light flashes on the wafer to completely dissociate the FeB pairs. Check the lifetime value in a single point after each flash session to ensure its saturation.

Task 3. Repeat the mapping measurement on the sample. How did the charge carrier lifetime change during the flashes? Calculate the iron constant of the system.

3.1.2 Investigation of Fe-B dissociation

Task 4. Load the measurable wafer (No.3) in the WT-2000 system, and repeat the iron concentration measurement used on the previous sample. Determine the iron concentration using the previously measured iron constant.

3.1.3 Determination of iron diffusion constant

Task 5. Load the first sample back into the system and perform lifetime mapping every 10 minutes. Determine the D_0 iron dissociation constant (Equation I.2.7) from the relaxation of $\tau(t)$.

Hint for the Lab Notes. Let us assume that the dissociation of FeB pairs is negligible and $\frac{\partial[FeB]}{\partial t} = R \cdot ([Fe]_{total} - [FeB])$.

3.2 Injection level dependent lifetime determination

3.2.1 Calibration tasks

In the following tasks you have to calibrate the eddy signal and the light intensity, which are the basics of the accurate ePCD measurement and evaluation.

Task 6. Measure the signal of the calibration wafers with known resistivity. The parameters of the calibration set can be found in the data sheet. Create the $V_{eddy} \sigma_{sh}$ function and fit the data points with a polynomial function.

Task 7. Repeat the previous task with thick silicon slugs. Calculate the apparent sheet conductance using the wafer calibration $\sigma_{sh,app}$. Define the sensitivity depth as follows:

$$d_{sense} = \rho \sigma_{sh,app}, \quad (II.3.1)$$

where ρ is the sample resistivity, given in the data sheet. Create the $V_{eddy} d_{sense}$ function and fit it with a suitable function.

Task 8. Measure the power of the excitation laser with a THORLABS light power measurement tool in a wide power range. Measure the diameter of the illumination window. Calculate the photon flux of the 1064 nm laser source.

3.2.2 Comparison of steady-state and PCD measurements on wafers and ingots

Task 9. Measure the given passivated silicon wafer with the ePCD measuring head. Use long enough pulses to achieve steady-state condition before the laser is terminated. Calculate the τ_{SS} value and the $\tau(\Delta n)$ curve from the measured signal using the results of the calibrations.

Task 10. Repeat the measurement with variable pulse energy at the same measurement point. Compare the results of the different methods to each other. How the measurement methods effect the $\tau(\Delta n)$ curve in case of passivated samples?

Task 11. Repeat the previous two tasks on an unpassivated silicon slug sample. Use d_{sense} as the sample thickness to calculate the injection level. What have you observed? Compare the results of the different methods to each other. How the surface

recombination effect the $\tau(\Delta n)$ curve in the different measurement methods?

3.2.3 Investigation of surface and bulk lifetime on Si ingots

Task 12. Determine the $\tau(\Delta n)$ curves of ingots with varying thickness!

Task 13. Determine the $\tau(\Delta n)$ curves of ingots with varying thickness. Separate the $\tau_{bulk}(\Delta n)$ and the $\tau_{surface}(\Delta n)$.

References

- [1] S. Sadhukhan, S. Acharya, T. Panda, N. C. Mandal, S. Bose, A. Nandi, G. Das, S. Maity, S. Chakraborty, P. Chaudhuri, and H. Saha, “Chapter 4 - evolution of high efficiency passivated emitter and rear contact (perc) solar cells,” in *Sustainable Developments by Artificial Intelligence and Machine Learning for Renewable Energies*, K. Kumar, R. S. Rao, O. Kaiwartya, M. S. Kaiser, and S. Padmanaban, Eds. Elsevier, 2022, pp. 63–129. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/B9780323912280000070>
- [2] S. Rein, *Lifetime Spectroscopy—A Method of Defect Characterization in Silicon for Photovoltaic Applications*. Springer Berlin, Heidelberg, 2005.
- [3] F. Dannhäuser, “Die abhängigkeit der trägerbeweglichkeit in silizium von der konzentration der freien ladungsträger—i,” *Solid-State Electronics*, vol. 15, no. 12, pp. 1371–1375, 1972. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/0038110172901311>
- [4] J. Krausse, “Die abhängigkeit der trägerbeweglichkeit in silizium von der konzentration der freien ladungsträger—ii,” *Solid-State Electronics*, vol. 15, no. 12, pp. 1377–1381, 1972. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/0038110172901323>
- [5] D.B.M. Klaassen, *A unified mobility model for device simulation—I. Model equations and concentration dependence*, *Solid-State Electronics*, 35 (1992), Pages 953-959. [https://doi.org/10.1016/0038-1101\(92\)90325-7](https://doi.org/10.1016/0038-1101(92)90325-7).
- [6] D.B.M. Klaassen, *A unified mobility model for device simulation—II. Temperature dependence of carrier mobility and lifetime*. *Solid-State Electronics*, Volume 35, Issue 7, 1992, Pages 961-967, [https://doi.org/10.1016/0038-1101\(92\)90326-8](https://doi.org/10.1016/0038-1101(92)90326-8).
- [7] Xiaodong Zhu, Xuegong Yu, Xiaoqiang Li, Peng Wang, Deren Yang, Quantification of characteristic parameters for the dissociation kinetics of iron–boron pairs in Czochralski silicon, *Scripta Materialia*, 2011, <https://doi.org/10.1016/j.scriptamat.2010.10.021>.
- [8] E.R. Weber, *Appl. Phys. A Mater. Sci. Process.* 30 (1983) 1